REPORT ABOUT: AUDIO-BASED DEPRESSION DETECTION

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ABSTRACT

This study uses real two-person counseling audio recordings to detect depression-related tasks (HRSD, Agitation, Retardation, Depression) and analyze emotion changes. Because real conversations have heavy noise and overlapping speech, we developed an automatic data cleaning pipeline. By using noise reduction, speech-to-text, and keyword matching, we successfully separated the interviewee's voice.

Based on this, the study conducted three main tasks: depression classification, emotion change difference (Diff) prediction, and emotion distribution analysis. The results show that deep learning models can effectively identify depression features in the audio of the Coping and Training stages. However, the models performed poorly in the EI stage classification task and the emotion change prediction task. In the emotion change prediction task, the Pearson Correlation Coefficient (PCC) of the model was very low, so the results lack reliability. Finally, the voice emotion distribution extracted by the model highly matches the doctors' clinical diagnoses.

Index Terms— Depression detection, ADHD detection, Audio process, Deep Learning

1. INTRODUCTION

Standard tests for depression mostly rely on subjective questionnaires filled out by patients. However, voice is an objective physical feature that is easier to collect and more real. Past studies on voice and depression mostly asked patients to read specific texts in a quiet room. But this study faces real conversations between two people. The difficulty of this situation is that the environment is noisy, and the doctor's and patient's voices are mixed together, with a lot of overlap and uncertainty.

To solve this problem, this study built a complete automatic system. It first extracts the patient's clean voice, and then lets the computer learn from it. This study explores how well the audio works on three tasks: First, judging if the patient has symptoms like depression, agitation, or retardation based on the extracted voice; Second, trying to predict the emotion score change (Diff) before and after the patient finishes a task; Third, exploring the differences in voice emotions (such as sadness, anger, and calmness) among patients with different symptoms. Through these tasks, this study hopes to find the useful range and limitations of voice data in real medical situations.

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2. MATERIALS & METHODS

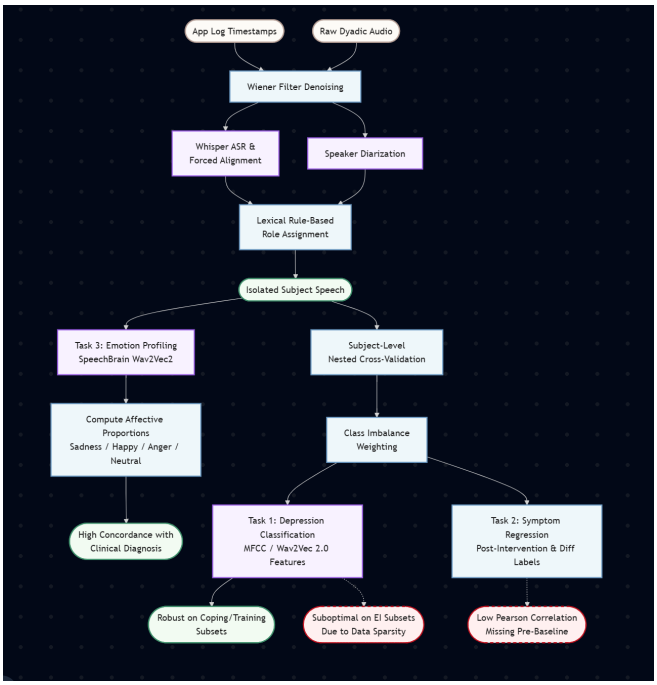

Fig. 1. Pipeline.

2.1. Preprocessing

To accurately extract the interviewee's voice from the two-person conversation:

First, the Wiener noise reduction algorithm was used to smooth and denoise the original audio signal. This helps to eliminate the interference of background noise on the following speech recognition as much as possible.

Next, the open-source Whisper model from OpenAI (specifically, the large-v3 version of faster-whisper) was used to convert the German audio into text.[1] This model not only keeps a high recognition accuracy in noisy environments, but

also outputs word-level timestamps. This provides a precise time basis for cutting the audio later.

To tell which sentences were spoken by the doctor and which by the patient, the speaker-diarization model from the ModelScope platform was used.[2] The estimated number of speakers was set to two. This model can divide the audio into separate speech parts based on different voice features.

Finally, a set of rules based on word statistics was designed to identify the real roles of the separated voices. By counting the frequency of keywords in each speaker's part, the person who often used words like "questionnaire (Fragebogen)", "emotion (Stimmung)", "scale (Skala)", and question sentences was judged as the interviewer. The person who frequently used the word "I (ich)" was judged as the interviewee. Through this process, the effective voice parts of the patient were successfully separated and saved. This provided clean data for the following model training.

2.2. Classification Model for Depression

To accurately classify the extracted patient voices, two different network architectures were compared in the experiment.

The first one is based on Mel-Frequency Cepstral Coefficients (MFCC) features.[3] It is good at capturing low-level acoustic features and voice textures. It uses a one-layer Bi-LSTM to extract temporal information from the audio for classification.[4]

The second one is a deep neural network based on the pre-trained Wav2Vec model.[5] Because this model was pre-trained on large audio datasets, it can better extract high-level hidden speech features and understand long-term context information in the audio.[6]

The classification task specifically covers four core clinical labels: whether the patient has depression (is_depression), whether they show agitation symptoms (is_agitation), whether they show retardation symptoms (is_retardation), and the Hamilton Rating Scale for Depression status (is_HRSD).

To prevent data leakage, a user-based dataset split was used (audio from the same user is not allowed to appear in the training set and validation set at the same time). To solve the problem of imbalanced data samples (positive samples are usually fewer), Class-Balanced weights were added to the loss function. By calculating the number of effective samples in each class, the system dynamically gives a higher error penalty to the minority class. This forces the model to pay more attention to the voice features of the sick group during training. Finally, a nested Cross-Validation (CV) strategy was used to ensure the robustness of the experiment. The inner loop is used to find the best learning rate, and the outer loop is used to evaluate the true generalization ability of the model [**?**].

2.3. Prediction of Emotional Change Differences

Although each audio segment includes scores for pre, post, and diff (on both Negative and Positive scales), the audio itself usually does not contain information from the "pre" stage. Therefore, this study only uses "post" and "diff" labels for prediction to explore feasibility.

To reduce the discrete nature and noise of the psychological scale scores, Min-Max normalization is applied to the Post labels. For the Diff labels, extreme value clipping and Z-score standardization are used. These steps help the model focus on the relative trends of emotion changes.

For the models, MFCC with Bi-LSTM and Wav2Vec are still used to extract features. Finally, a regression head is used to predict the scores.

2.4. Emotion Extraction

For this task, this study uses a speech emotion recognition model, SpeechBrain , which was pre-trained on public datasets, to classify the extracted audio recordings.[7] This study calculates the proportions of different emotion categories, such as "sad," "happy," "neutral," and "angry," shown by patients in the conversations. Finally, these results are compared with the actual medical diagnosis labels for analysis.

3. EXPERIMENTS & RESULTS

For classification tasks, Accuracy and F1-score are mainly used as evaluation metrics. For regression tasks, Mean Squared Error (MSE), Mean Absolute Error (MAE), and Pearson Correlation Coefficient (PCC) are introduced to measure the correlation between predicted and real values. In the data splitting stage, a user-based nested cross-validation is strictly applied to ensure that the experimental results are not affected by data leakage.

3.1. Experiment on Multi-Label Classification of Depression

Nested CV Compare (Mean Acc)

Label	('Coping', 'MFCC')	('Coping', 'Wav2Vec')	('Training', 'MFCC')	('Training', 'Wav2Vec')
is_HRSD	0.6128	0.6651	0.7464	0.6562
is_agitation	0.6441	0.7033	0.642	0.7313
is_depression	0.5046	0.6226	0.5261	0.5942
is_retardation	0.747	0.7726	0.7624	0.7978

Nested CV Compare (F1 Score)

Label	('Coping', 'MFCC')	('Coping', 'Wav2Vec')	('Training', 'MFCC')	('Training', 'Wav2Vec')
is_HRSD	0.5573 ± 0.0812	0.5511 ± 0.0685	0.6557 ± 0.0731	0.5719 ± 0.0935
is_agitation	0.6053 ± 0.0484	0.6258 ± 0.0393	0.5865 ± 0.0850	0.6509 ± 0.1086
is_depression	0.4840 ± 0.0427	0.6092 ± 0.0399	0.5136 ± 0.0356	0.5910 ± 0.0229
is_retardation	0.5814 ± 0.0854	0.5605 ± 0.0708	0.5788 ± 0.0853	0.5089 ± 0.0523

Fig. 2. Comparison Between MFCC and Wav2Vec on Multi-label Classification Tasks.

Model Performance Comparison and Data Analysis: As can be directly seen from Fig.2, the deep learning-based Wav2Vec model showed better classification performance than the traditional MFCC model on the audio data of both the Coping and Training stages. Specifically, the Wav2Vec model almost entirely leads in both Accuracy and F1-score across all four labels: "is_depression", "HRSD", "agitation (is_agitation)", and "retardation (is_retardation)". For example, in the depression classification of the Coping stage, the F1-score of Wav2Vec reached 0.6732, which is clearly higher than MFCC's 0.5750. In addition, in the Training stage, Wav2Vec achieved a highest accuracy of 0.8351 on the "retardation" label. These specific data improvements strongly show that the deep voice features extracted by the pre-trained model can more accurately reflect the voice characteristics of depressed interviewee than simple low-level spectrum features.

EI Task Failure Analysis: This study found that the classification task results in the EI stage were very poor, and effective classification could not be done. After a deep analysis of the process, it was found that the extraction of EI audio was based on the time ranges recorded in the mobile phone logs (app_logs). However, there is a clear mismatch between the times recorded in these logs and the actual time the interviewee spoke. After the automatic transcription and speaker separation process, many set time ranges actually had no content or it was not the patient speaking. This caused the amount of useful audio data in the EI stage to drop sharply, which could not support the model for normal training and prediction.

3.2. Regression Analysis of Emotional Change Differences (Diff)

Model	Stage	Target	MAE (Mean±Std)	MSE (Mean±Std)	PCC (Mean±Std)
mfcc	diff	n	0.7700 ± 0.0565	1.1590 ± 0.2133	0.1398 ± 0.0573
		p	0.8462 ± 0.0899	1.2163 ± 0.2445	0.0788 ± 0.1024
	post	n	2.1693 ± 0.2021	7.4239 ± 1.4764	0.1531 ± 0.1575
		p	2.2365 ± 0.2762	7.5235 ± 1.8225	0.1007 ± 0.1043
wav2vec	diff	n	0.7618 ± 0.0763	1.1389 ± 0.1887	0.0044 ± 0.0317
		p	0.8499 ± 0.1011	1.1766 ± 0.2977	0.0305 ± 0.0628
	post	n	2.1679 ± 0.2292	6.9473 ± 1.3996	0.1322 ± 0.0752
		p	2.0704 ± 0.2611	6.6952 ± 1.4505	0.0206 ± 0.1390

Fig. 3. Results of Emotional Change Differences.

As can be seen from Fig.3, although the Mean Squared Error (MSE) or Mean Absolute Error (MAE) calculated by the model are relatively low, the key metric for measuring the change trend—the Pearson Correlation Coefficient (PCC)—is very low. A very low PCC means that the score change trend predicted by the model has almost no relationship with the interviewee's real emotional change. This proves that a single audio segment from the current dataset is not suitable for the task of predicting the "change difference". The final prediction results lack clinical credibility.

3.3. Exploratory Analysis of the Distribution of Tonal Emotions

By analyzing the voice emotions of different clinical symptom groups, this study reached conclusions that highly match common medical knowledge:

- Depression: Subjects with depression showed a higher proportion of "sad" and "neutral" emotions in the conversations.
- Agitation: Patients diagnosed with agitation had a clearly higher proportion of high-arousal emotions like "angry" in their audio compared to other groups.
- Retardation: The emotional changes of these patients were very small. The statistical charts show that their emotions were almost entirely made up of "neutral" and "sad," and rarely showed fluctuations like "happy" or "angry."
- HRSD: Patients with high HRSD scores also showed a clear polarization of negative emotions. These results directly prove that the voice features extracted by the process of this study can truly reflect the patients' psychological and pathological states.

Detailed visualization charts are provided in the appendix.

4. CONCLUSION

The conclusions of this study are divided into two parts: system implementation and emotion distribution, and the verification and exploration of failed tasks.

4.1. System Implementation and Emotion Distribution

First, facing the noisy two-person conversation scenario, this study successfully built an audio extraction system based on transcription. Through noise reduction, speech recognition, and specific keyword rule matching, this system can effectively separate the patient's voice from complex conversations.

Second, based on the separated clean audio, this study compared two classification models, Wav2Vec and MFCC, on the data of the Coping and Training stages. The experimental results prove that the deep learning model Wav2Vec can effectively identify depression features. Its accuracy and F1-score on labels such as depression, agitation, retardation, and HRSD are clearly better than the traditional MFCC model.

Finally, in the emotion analysis task, the emotion distribution extracted in this study highly matches the doctors' clinical diagnoses. For example, patients in the depression and retardation groups had a very high proportion of "sad" and

”neutral” emotions in their voices, while patients in the agitation group showed a higher proportion of high-arousal emotions like ”angry”. This proves that the emotion features extracted from audio have real medical reference value.

4.2. Failure Verification and Future Suggestions

Besides the successful classification tasks, this study also objectively recorded and analyzed two failed exploration tasks. This provides an important reference for building future medical datasets:

First, the prediction task of the emotion change difference (Diff) failed. Although the error values are not high, the change trend (PCC) predicted by the model is very low, and the results lack credibility. The reason is the inherent flaw of the dataset: the audio data only covers the intervention process and lacks the baseline voice before the task. At the same time, the tiny emotional changes of a person within a few minutes are not obvious in the physical features of the voice.

Second, the audio classification task in the EI stage also failed. The main reason is that there is a serious mismatch between the time periods recorded by the external mobile phone logs and the actual time the patients spoke. After the strict voice extraction process, the massive loss of useful audio made it impossible for the model to train.

Future Suggestions: Based on the above explorations, this study suggests that in future medical diagnosis systems, predicting tiny emotional changes (Diff) over a short time may not be achieved simply by relying on the audio modality. It needs to be combined with multimodal data such as facial expressions or physiological indicators. For the daily audio extraction in the EI stage, we should stop relying on the time logs of external Apps, and instead use manual re-labeling of the valid audio ranges or introduce more robust Voice Activity Detection (VAD) technology for time alignment.

5. REFERENCES

[1] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever, “Robust speech recognition via large-scale weak supervision,” *arXiv preprint arXiv:2212.04356*, 2022.

[2] Alibaba DAMO Academy, “Speech cam++ speaker diarization common model,” `https://www.modelscope.cn/models/iic/speech_campplus_speaker-diarization_common/`, 2023, Accessed: 2026-03-21.

[3] Z. K. Abdul and A. K. Al-Talabani, “Mel frequency cepstral coefficient and its applications: A review,” *IEEE Access*, vol. 10, pp. 122136–122158, 2022.

[4] Zhiheng Huang, Wei Xu, and Kai Yu, “Bidirectional lstm-crf models for sequence tagging,” *arXiv preprint arXiv:1508.01991*, 2015.

[5] Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020, vol. 33, pp. 12449–12460.

[6] Xiangsheng Huang, Fang Wang, Yuan Gao, Yilong Liao, Wenjing Zhang, Li Zhang, and Zhenrong Xu, “Depression recognition using voice-based pre-training model,” *Scientific Reports*, vol. 14, no. 1, pp. 13322, 2024.

[7] Mirco Ravanelli, Titouan Parcollet, Peter Plantinga, Aku Rouhe, Samuele Cornell, Loren Lugosch, Cem Subakan, Nauman Dawalatabad, Abdel Heba, Jianyuan Zhong, et al., “Speechbrain: A general-purpose speech toolkit,” *arXiv preprint arXiv:2106.04624*, 2021, Code available at: `https://github.com/speechbrain/speechbrain`.

A. APPENDIX: COPING STAGE ANALYSIS

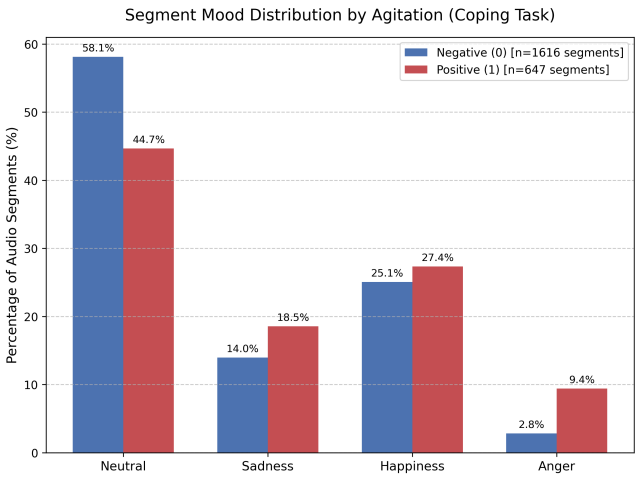

Fig. A1. Mood distribution in Is_agitation.

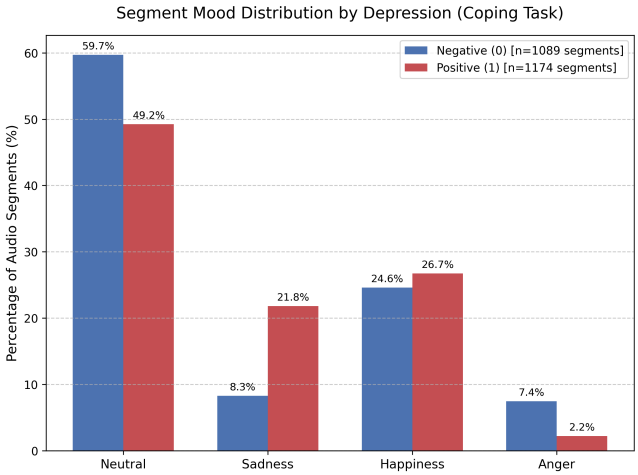

Fig. A2. Mood distribution in Is_depression.

B. APPENDIX: TRAINING STAGE ANALYSIS

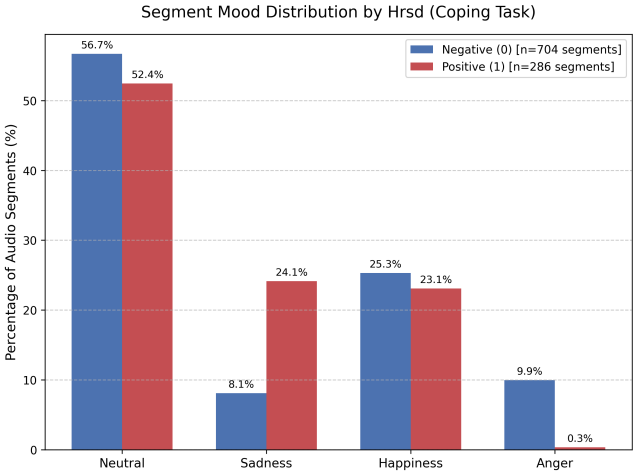

Fig. A3. Mood distribution in Is_HRSD.

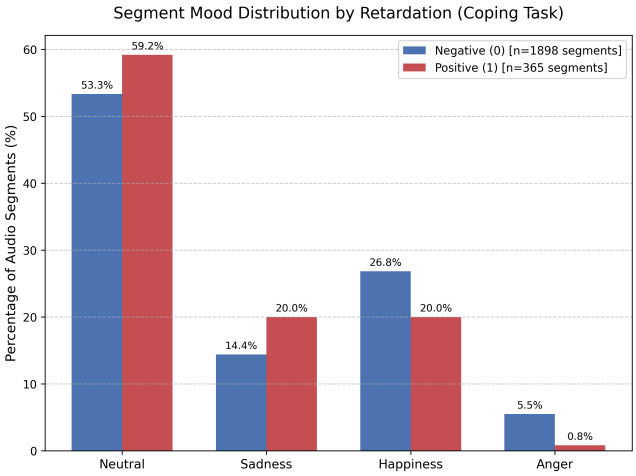

Fig. A4. Mood distribution in Is_retardation.

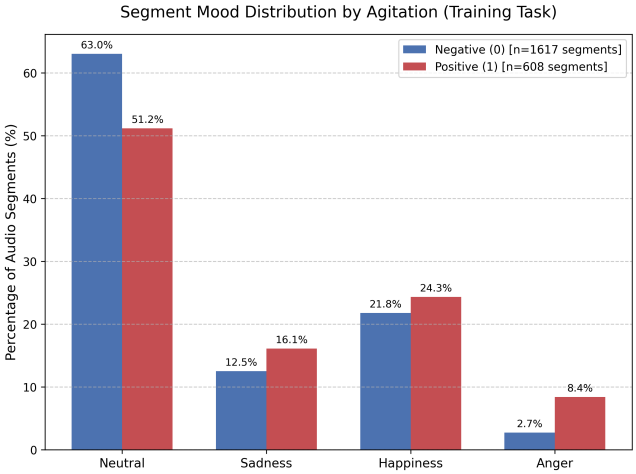

Fig. A5. Mood distribution in Is_agitation.

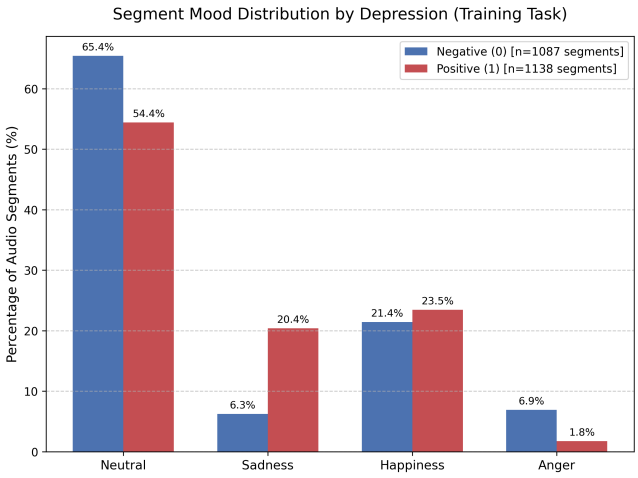

Fig. A6. Mood distribution in Is_depression.

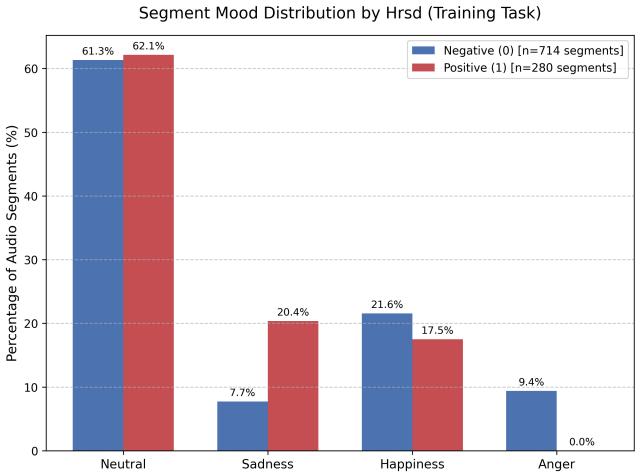

Fig. A7. Mood distribution in Is_HRSD.

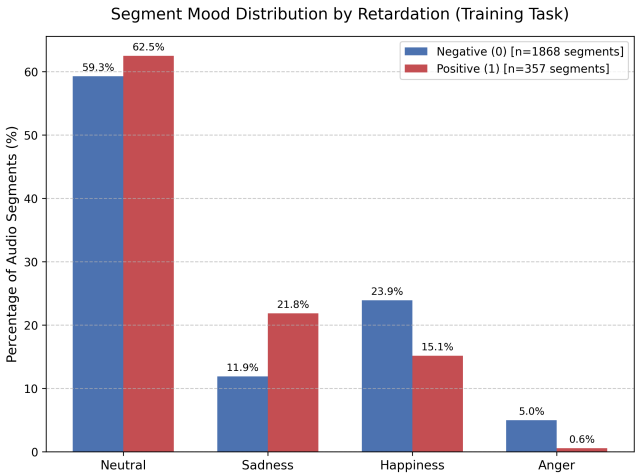

Fig. A8. Mood distribution in Is_retardation.